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# Gun Possession & Regression

*Understanding Household Firearm Rate through  
Feature Importance*

# The Problem

- **NRA:** Gun control does not work
  - *Criminals don't obey gun control laws, background checks aren't effective, crime goes down when good guys have guns<sup>1</sup>*
- **Everytown:** Gun safety policies save lives
  - *In states where elected officials have taken action to pass gun safety laws, fewer people die by gun violence<sup>2</sup>*
- **Pew Research:** About three-quarters (76%) of Democrats say the ease with which people can legally obtain guns contributes a great deal or fair amount to gun violence, compared with just 39% of Republicans<sup>3</sup>

1. "Why Gun Control Doesn't Work," NRA-ILA

2. "Gun Safety Policies Save Lives," Everytown Research & Policy, March 11, 2026

3. J. Baxter Oliphant, "Bipartisan Support for Some Gun Proposals, Stark Partisan Divisions on Many Others," Pew Research Center, June 23, 2017

# My Question

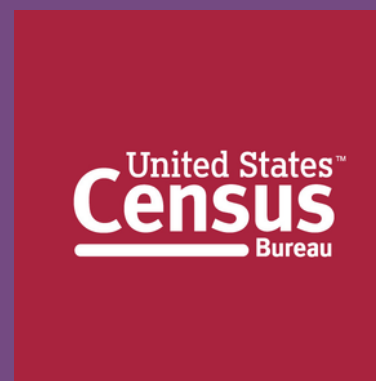
*Why do Americans own guns?*

# The Initial Data

- **Personal Income**
- **Total Employment**
- **Disposable Personal Income**
- **Personal Consumption Expenditures<sup>1</sup>**



- **Public Assistance Income or Food Stamps/SNAP in the Past 12 Months for Households<sup>2</sup>**
- **Educational Attainment<sup>3</sup>**
  - %  $\geq$  HS
  - %  $\geq$  Bachelor's
  - Pov for 25yo $\leq$ , <HS
  - Pov for 25 $\leq$ ,  $\geq$ Bachelor's



- **Household firearm rate<sup>4</sup> (target)**



1. U.S. Bureau of Economic Analysis, "SASUMMARY State annual summary statistics: personal income, GDP, consumer spending, price indexes, and employment"

2. U.S. Census Bureau, "Public Assistance Income or Food Stamps/SNAP in the Past 12 Months for Households," American Community Survey, ACS 1-Year Estimates Detailed Tables, Table B19058

3. U.S. Census Bureau, "Educational Attainment," American Community Survey, ACS 1-Year Estimates Subject Tables, Table S1501

4. Schell, Terry L., Samuel Peterson, Brian G. Vegetabile, Adam Scherling, Rosanna Smart, and Andrew R. Morral, State-Level Estimates of Household Firearm Ownership. Santa Monica, CA: RAND Corporation, 2020.

# Cleaning + Preparing Data

*US state data from 2010-2016  
(350 instances)*

## **Columns:**

LocYear  
% >= HS  
% >= Bachelor's  
Pov for 25yo<=, <HS  
Pov for 25<=, >=Bachelor's  
SNAP\_rate  
DPI\_rate  
PI\_rate  
HFR

**Making Data Comparable**

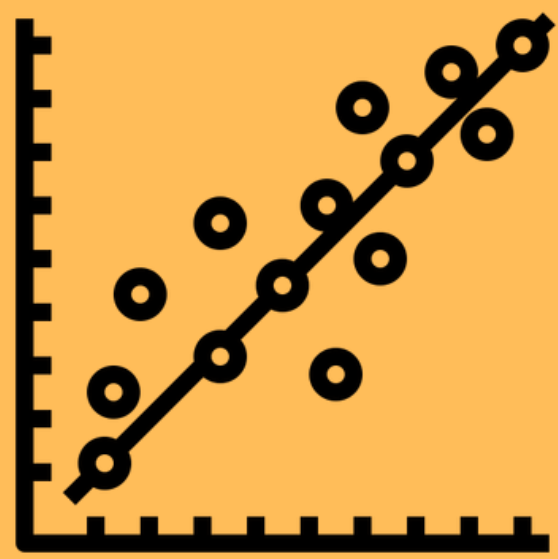
$\$ \rightarrow \%$

**Dropping Unneeded Columns**

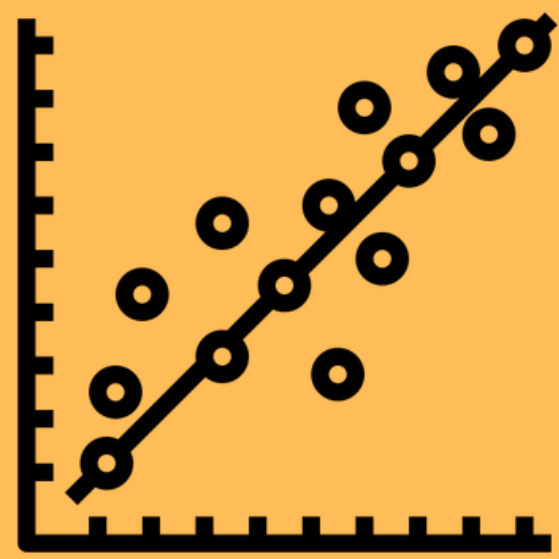
**Creating Universal Index**

*AL2010, WV2016*





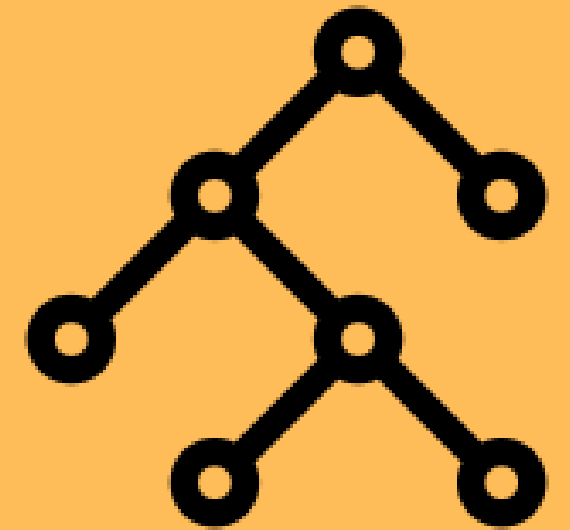
**Linear  
Regression**



**Single Linear  
Regression**



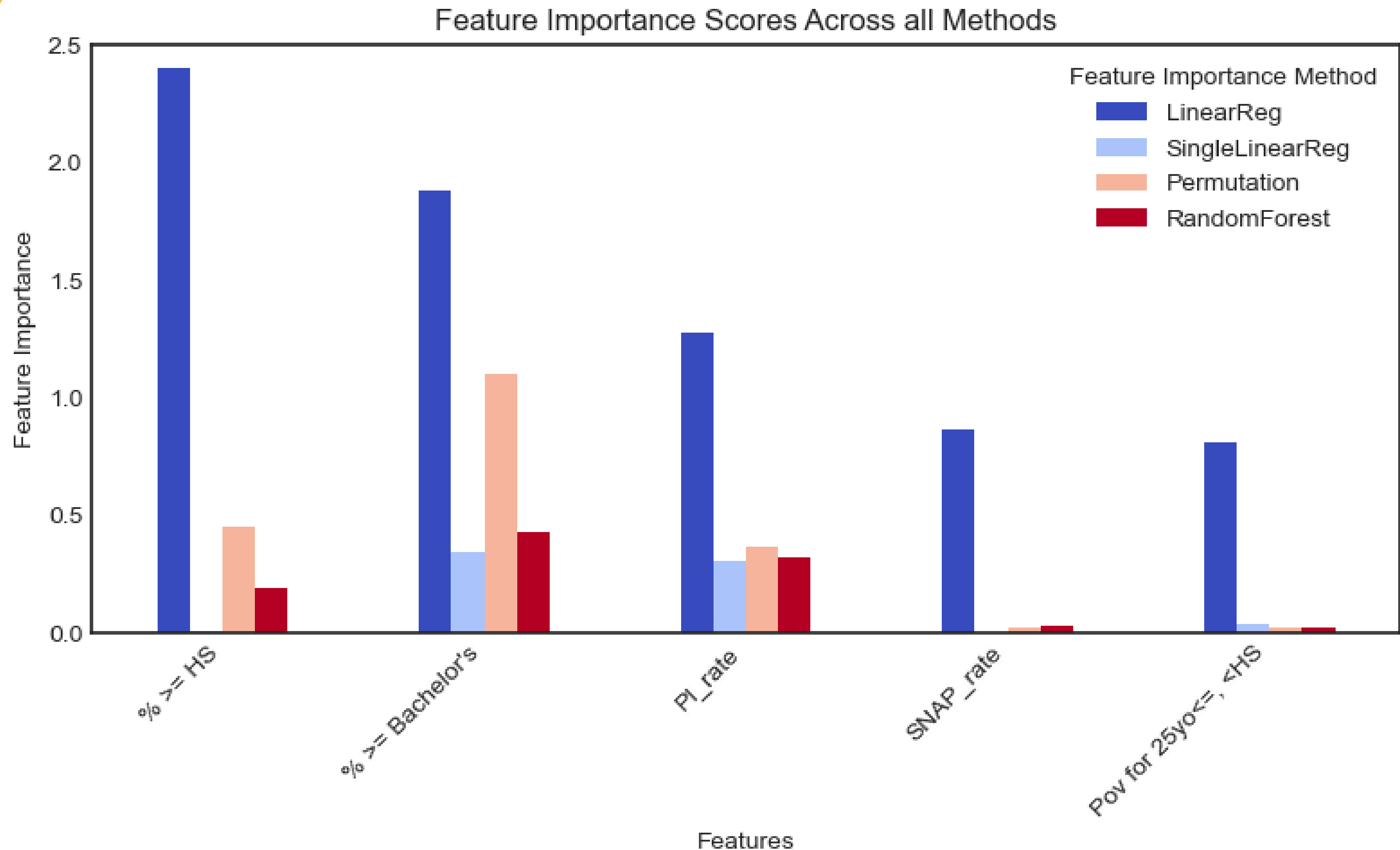
**Permutation**



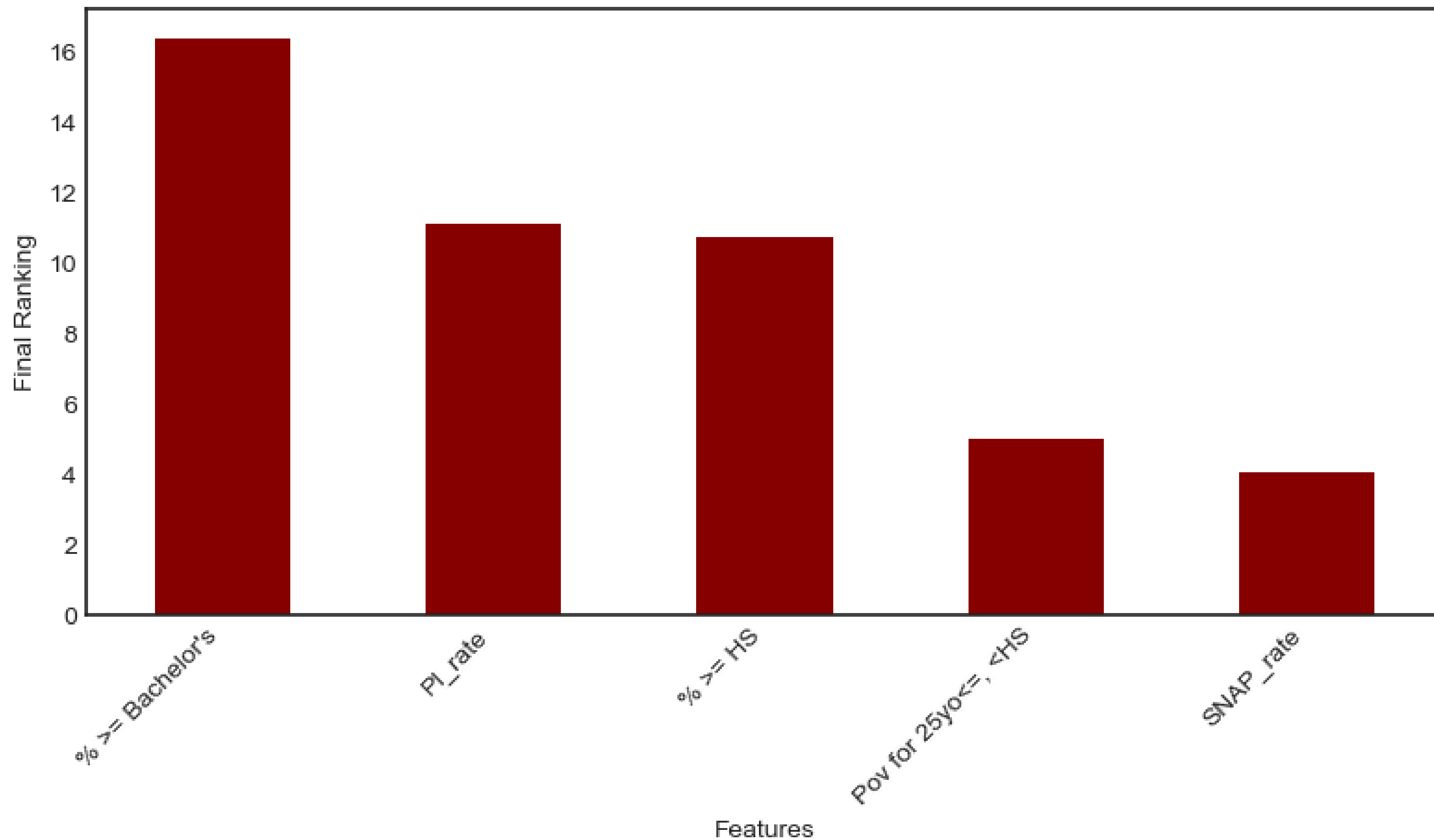
**Random  
Forest**

# Methods

**Feature importance** involves calculating the score for all input features in a machine learning model to establish the importance of each feature in the decision-making process. **The higher the score for a feature, the larger the effect it has on the model to predict a certain variable.<sup>1</sup>**



Feature Importance Scores Ranking





# Calling their Bluff (?)

*What features are possibly better predictors of household firearm rate?*

## Previous Model Scores ( $r^2$ )

- **Linear Regression:** 0.5919085723335016
- **Permutation:** 0.5974127798037477
- **Random Forest:** 0.614404866142719

*About seven-in-ten gun owners (72%) say protection is a major reason they own a gun<sup>1</sup>*

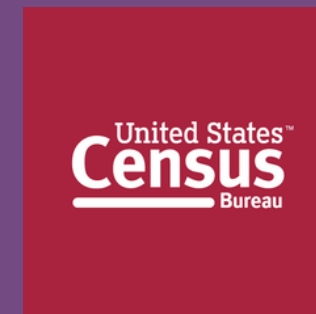
***Republican** gun owners are **more likely** than **Democratic** owners to say owning one gives them feelings of **safety** and **enjoyment**, while **Democratic** owners are **more likely** to say they **worry** about having a gun in the home<sup>1</sup>*

# The Final Data

- Robbery
- Aggravated assault
- Property crime
- Burglary
  - all per 100,000 people<sup>1</sup>



- House Democrats
- House Republicans<sup>2</sup>
  - House Leaning
- Educational Attainment
  - %  $\geq$  Bachelor's



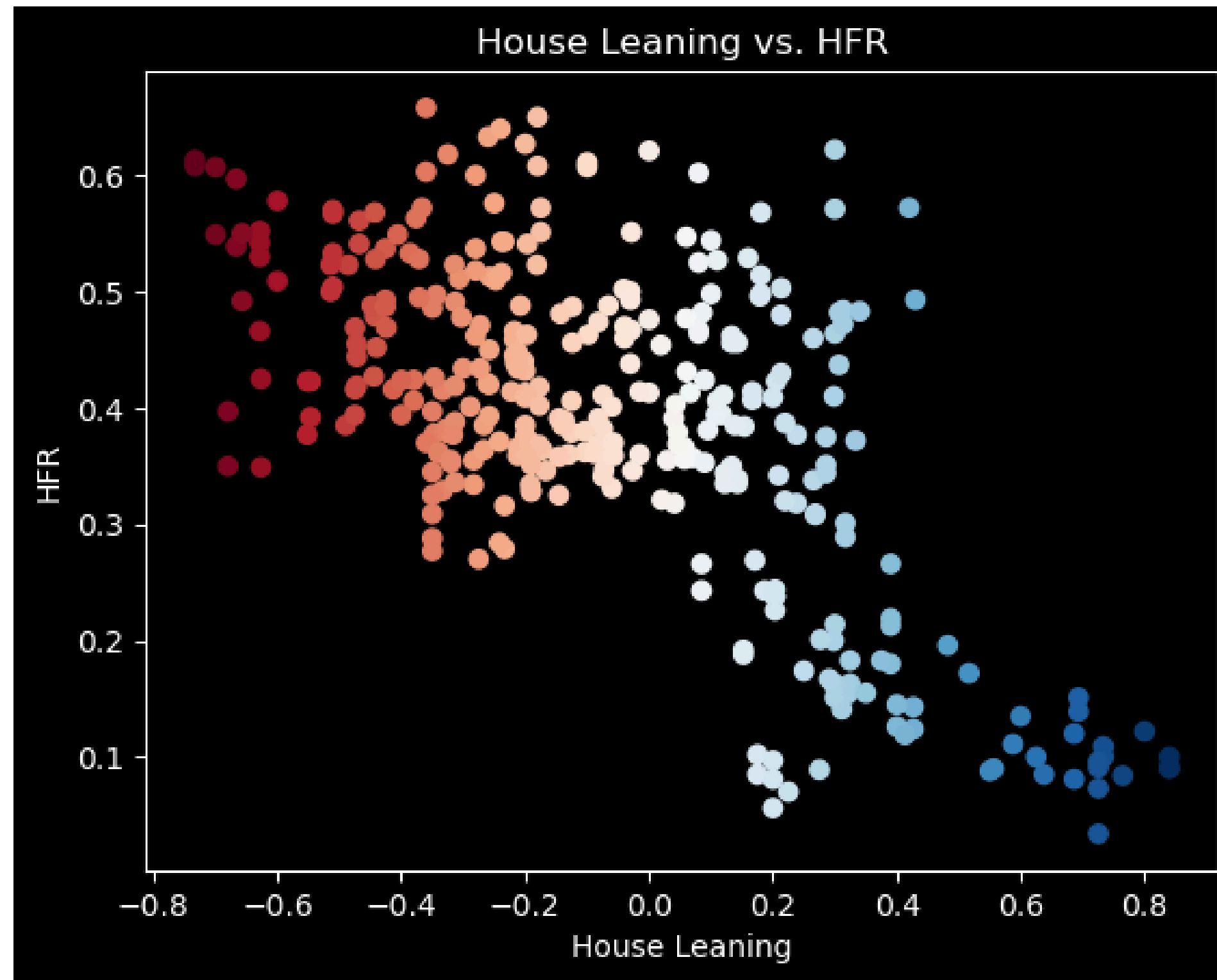
- Personal Income Rate
- Household firearm rate (target)



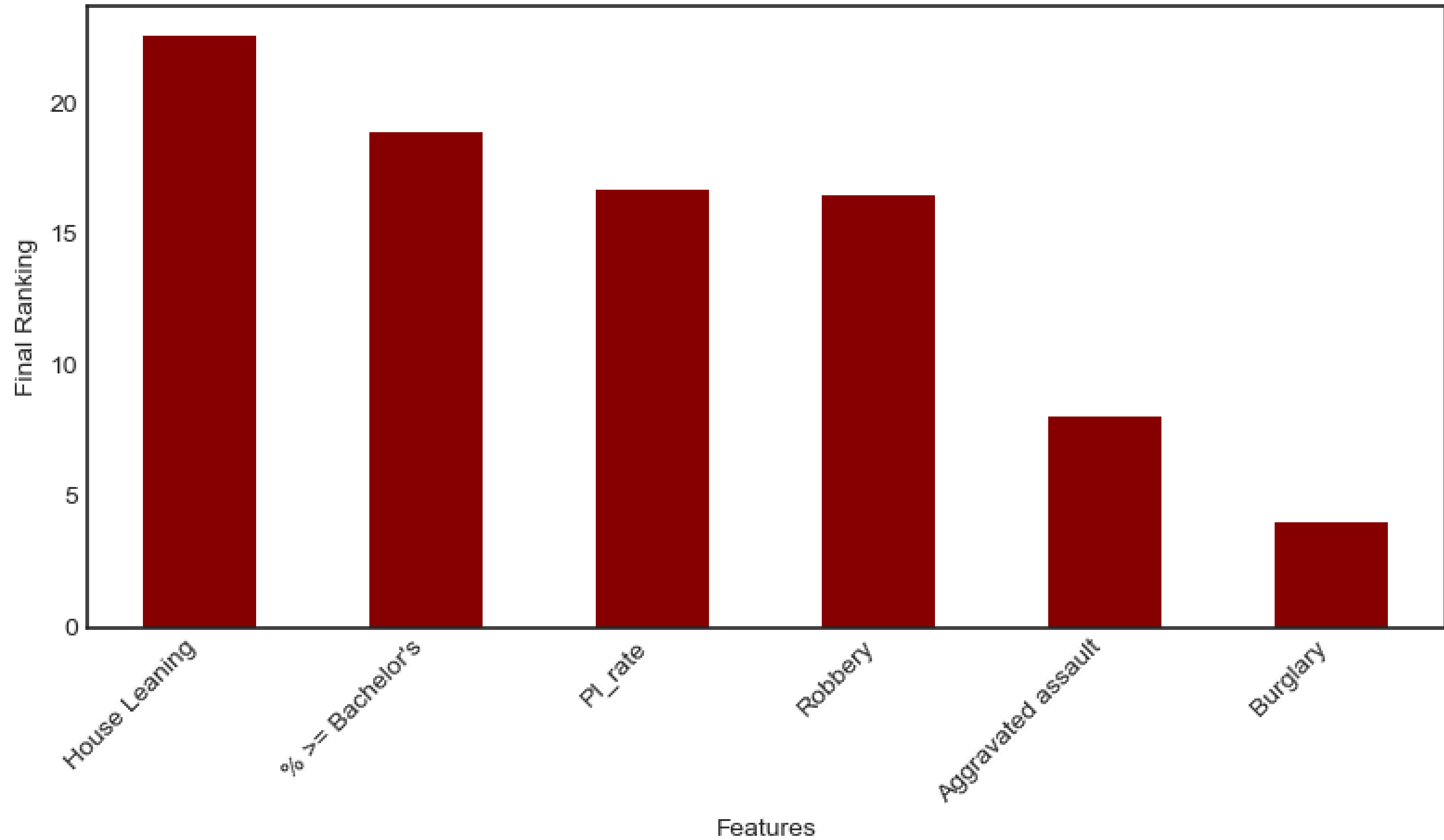
1. Crime/Law Enforcement Stats (UCR Program) | Federal Bureau of Investigation," FBI

2. "State Partisan Composition Database," National Conference of State Legislatures

# EDA Results



Feature Importance Scores Rankings w/ Political and Revised Crime Data



# Results + Significance

- **Linear Regression:** 0.73478
  - *Prev:* 0.5919 (+0.143)
- **Permutation:** 0.87354
  - *Prev:* 0.5974 (+0.276)
- **Random Forest:** 0.77912
  - *Prev:* 0.6144 (+0.165)

Notable model improvement seen with added political and crime data

- **House Leaning is #1 ranked feature**
  - Followed by %  $\geq$  Bachelor's
- Crime data has the three lowest rankings of the final dataset
  - Self-defense argument

*Is this the full picture?*



# Limitations

## Data

- Data from different sources
- Outdated (2010-2016)
- Comparability/representability

## Methods

- Linear Regression feature importance
- Comparing importance scores

## Analysis

- Confounding variables



# The End!

*To see the code for this project, navigate to [SMUREU/miniproj\\_SMUREU](#)*

“Crime/Law Enforcement Stats (UCR Program) | Federal Bureau of Investigation.” FBI. Accessed June 11, 2026. <https://www.fbi.gov/how-we-can-help-you/more-fbi-services-and-information/ucr>.

“Gun Safety Policies Save Lives.” Everytown Research & Policy, March 11, 2026. <https://everytownresearch.org/rankings/>.

Oliphant, J. Baxter. “Bipartisan Support for Some Gun Proposals, Stark Partisan Divisions on Many Others.” Pew Research Center, June 23, 2017. <https://www.pewresearch.org/short-reads/2017/06/23/bipartisan-support-for-some-gun-proposals-stark-partisan-divisions-on-many-others/>.

Schaeffer, Katherine. “Key Facts about Americans and Guns.” Pew Research Center, July 24, 2024. <https://www.pewresearch.org/short-reads/2024/07/24/key-facts-about-americans-and-guns/>.

Schell, Terry L., Samuel Peterson, Brian G. Vegetabile, Adam Scherling, Rosanna Smart, and Andrew R. Morral, State-Level Estimates of Household Firearm Ownership. Santa Monica, CA: RAND Corporation, 2020. <https://www.rand.org/pubs/tools/TL354.html>.

Shin, Terence. “Understanding Feature Importance in Machine Learning.” Built In, November 7, 2024. <https://builtin.com/data-science/feature-importance>.

“State Partisan Composition Database.” National Conference of State Legislatures. Accessed June 11, 2026. <https://app.powerbi.com/view?r=eyJrIjoiazZWQ5NDAwYzMtOWMwNC00ZDBkLTg3YzgtYjQxMGlxMDFmN2E4IiwidCI6IjM4MmZiOGIwLTRkYzMtNDEwNy04MGJkLTM1OTViMjQzMmZhZSIsImMiOjZ9>.

U.S. Bureau of Economic Analysis, "SASUMMARY State annual summary statistics: personal income, GDP, consumer spending, price indexes, and employment" (accessed Thursday, June 4, 2026)

U.S. Census Bureau, "Educational Attainment," American Community Survey, ACS 1-Year Estimates Subject Tables, Table S1501, accessed on June 11, 2026, <https://data.census.gov/table/ACSST1Y2024.S1501?q=S1501>.

U.S. Census Bureau, "Public Assistance Income or Food Stamps/SNAP in the Past 12 Months for Households," American Community Survey, ACS 1-Year Estimates Detailed Tables, Table B19058, accessed on June 11, 2026, <https://data.census.gov/table/ACSST1Y2024.B19058?q=B19058>.

“Why Gun Control Doesn’t Work.” NRA-ILA. Accessed June 11, 2026. <https://www.nraila.org/why-gun-control-doesn-t-work/>.